Host forensics and examination approaches II

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1. Windows OS
2. Registry Analysis
3. Log Analysis
4. Encryption
5. Timeline

Learning Outcomes

1. Develop an understanding on the main areas a forensic analysis of Windows OS can concentrate
2. Discuss the main Registry Hives that may need to be examined during an investigation
3. Review the analysis of Windows OS logs
4. Discuss the Windows Encryption mechanisms
5. Understand the importance of developing a timeline for an investigation

Analysis Categories

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Physical Media Analysis

- a custom storage layout and contents to a standard interface, e.g. IDE, SCSI
- the bytes of the media
- processing the custom layout , recovering deleted data

Media Management Analysis

- the organisation of the storage media.
- another collection of bytes from the media.
- E.g. dividing a hard disk into partitions, organizing multiple disks into a volume, and integrating multiple memory chips into memory space.

File System Analysis

- translate the bytes and sectors of the partition to directories and files.
- the actual file content.
- includes viewing directory and file contents and recovering deleted files.

Application Analysis

- translates data, typically returned from the file system, into the custom format needed by the application.
- viewing log files, configuration files, images, documents and reverse engineering executables.
- The input data will typically come from the file system
- applications such as databases may read directly from the disk.
- Analysis of logs generated by network services, e.g. firewall or web server

Network Analysis

- translates the lowest level data from a physical network or wireless network to the data that is used by an application.
- Analyzing network packets and IDS alerts.

Memory Analysis

- translates the bytes of the memory media to processes and system data.
- identifying the code that a process was running
- extracting sensitive data that was not stored elsewhere.

Windows Forensic Analysis

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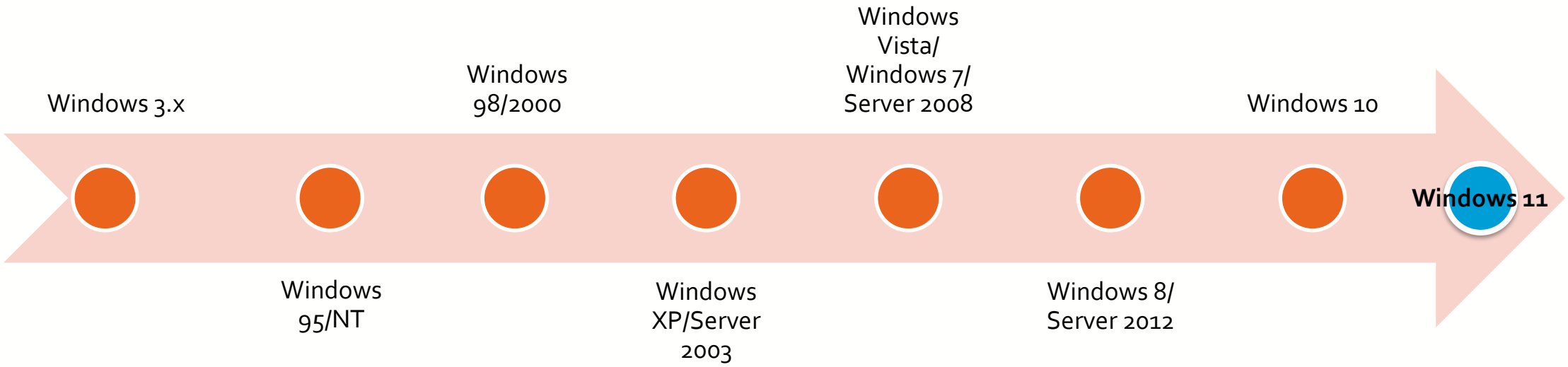
Windows 3.x
Windows 95/NT
Windows 98/2000
Windows XP/Server 2003
Windows Vista/ Windows 7/ Server 2008
Windows 8/ Server 2012
Windows 10
Windows 11

Windows Details

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32-bit

- Addresses up to 4,294,967,295 bytes
- Limited to 4 gigabytes (GB) of RAM
- Older architecture, now largely obsolete
- Limited modern security support

64-bit

- Addresses up to 18,446,744,073,709,551,616 bytes
- Referred to as x64
- Standard for modern systems

Why architecture matters in forensics

- Memory acquisition and analysis differ
- Registry redirection occurs on 64-bit systems
- Program execution artefacts may exist in parallel locations
- Tool compatibility depends on OS architecture

Windows Forensics Overview

- Focuses on analysis of Windows operating systems
- Examines user activity, system behaviour, and artefacts
- Common in corporate and criminal investigations
  - Key Evidence Sources in Windows
    - File systems (NTFS)
    - Registry hives
    - Event logs
    - User profile data

Challenges in Windows Forensics

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Technical challenges

- Volume and complexity of artefacts
- Multiple overlapping data sources
- Conflicting timestamps and records
- 32-bit vs 64-bit redirection

System behaviour challenges

- Background OS activity
- Automated updates and services
- User vs system-initiated actions

Security and protection challenges

- Full-disk and file-level encryption
- Secure boot and hardware protection
- Restricted access to memory and storage

Operational challenges

- Live vs dead-box acquisition decisions
- Evidence volatility
- Tool compatibility and version drift

Windows Registry Analysis

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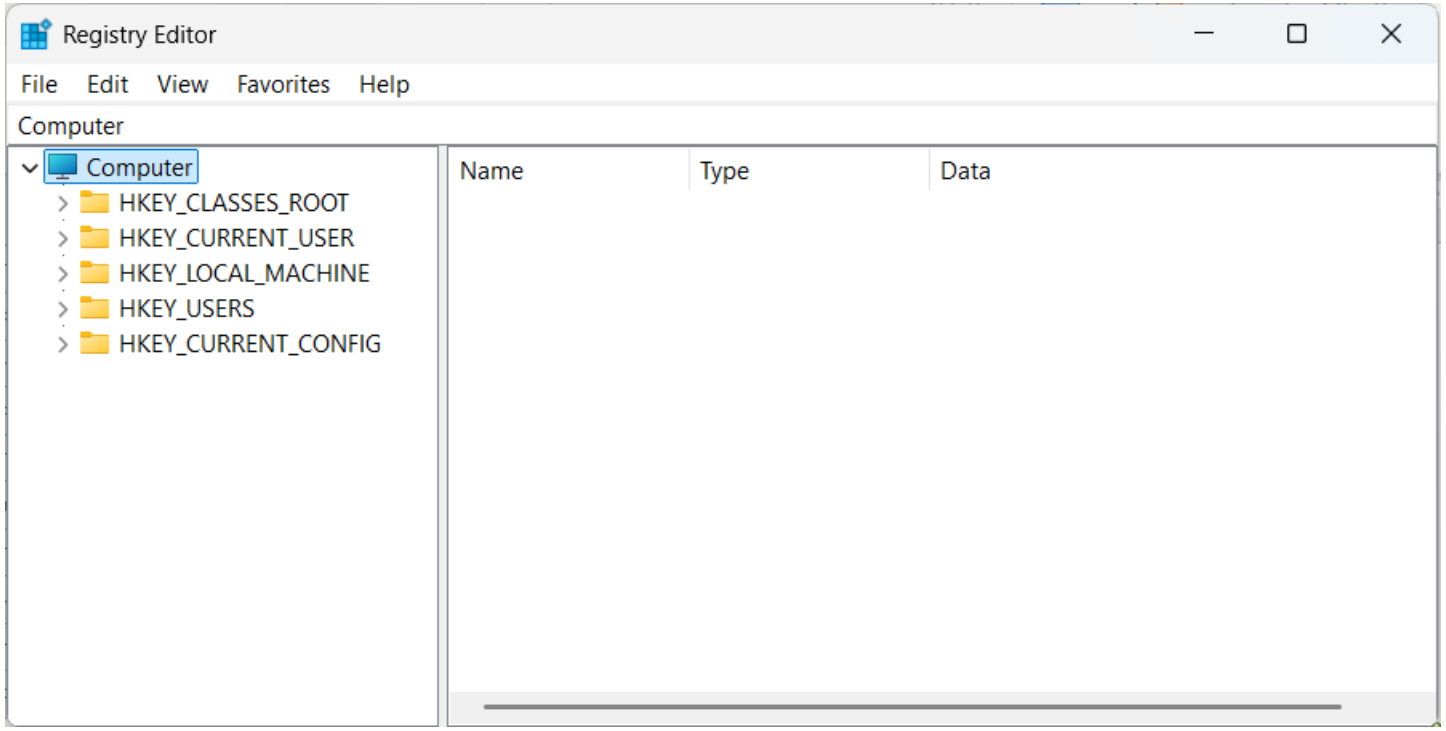
Purpose of the Registry

Windows OSs since Win95 have used a binary database to record
- details on the OS and other programs
- data concerning user actions and preferences,
- the current configuration of the system and its components

- The registry is available whenever the system is running
  - programs that run intermittently can keep persistent data in the registry.
  - information is loaded at every system boot and remains resident until shutdown.
  - programs store all types of data in registry entries.
    - they use standard Win32 application programming interfaces (APIs) to retrieve data from the registry.
    - the program specifies an entry by its path and name and the API returns the value of the entry.
    - programs also use standard APIs to add and delete registry content and to change the values of registry entries.

The Registry Hives

- a set of discrete files
  - Each hive contains a Registry tree, which has a key that serves as the root (HKEY_Name) of the tree
  - Subkeys and their values reside beneath the root
  - Stored on disk, loaded into memory at runtime



Core Registry Hives in Forensics

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SYSTEM

hardware, services, time zone

SOFTWARE

OS and application configuration

SAM

user accounts and credentials (restricted)

NTUSER.DAT

user-specific activity and settings

Forensics & the Registry

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System configuration

- Computer name and system identifiers
- Installed operating system version and build
- Time zone and regional settings
- Installed hardware and drivers

User accounts and activity

- User account names and SIDs
- Last login information (where available)
- User profile paths
- Recently accessed resources

Program execution

- Installed applications
- Recently executed programs
- Startup programs and persistence mechanisms

Device and peripheral usage

- USB storage devices connected to the system
- Device serial numbers and first/last connection times
- Mounted drives and volumes

Network configuration

- Network interfaces and configurations
- Previously connected networks
- Wireless network identifiers (SSIDs)

The methodology

1. ***Locate and Extract the Registry Hives***
   - The version of the Operating System will determine which files are present.
2. ***Examine the Hives***
   - The binary files require specific tools to access:
     - Regedit (Supplied with Windows).
     - Registrar Registry Manager 7 by Resplendence Software,
     - Linux Regviewer,
     - EnCase,
     - Registry Viewer from AccessData.
3. ***Find Items Relating to the user and his actions in relation to the investigation***
   - Information may be in different places on different systems, assistance can be found from:
     - Microsoft's Technet – Inside the Registry (Microsoft, 2012)
     - Registry Guide for Windows (http://www.pctools.com/guides/registry/)
     - Useful tool: Windows Registry Recovery Tool

Forensics & the Registry

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Examples

- Wireless Network SSID
  - HKEY_LOCAL_MACHINE \SOFTWARE\Microsoft\WZCSVC\Parameters\Interfaces
- List of all the Wi-Fi networks
  - HKLM\SOFTWARE \Microsoft\WindowsNT\CurrentVersion\NetworkList\Profiles\
  - Includes SSID and date the computer first connected to the network
- MAC address of the wireless access point
  - HKLM\SOFTWARE\Microsoft\WindowsNT\CurrentVersion\NetworkList\Signatures\Unmanaged\{ProfileGUID}
- List of mounted volumes
  - HKEY_LOCAL_MACHINE \SYSTEM\MountedDevices

Windows Security and Relative ID

The Windows Registry utilizes an alphanumeric combination to uniquely identify a security principal or security group.

- Security ID (SID)
  - used to identify the computer system.
- Relative ID (RID)
  - used to identity the specific user on the computer system.
- The SID appears as:
  - S-1-5-21-927890586-3685698554-67682326-1005

Well-Known SIDs

- Everyone (S-1-1-0):
  - automatically includes everyone who uses the computer, even anonymous guests.
- Creator Owner (S-1-3-0):
  - A placeholder in an inheritable access control entry (ACE) When the ACE is inherited, the system replaces this SID with the SID for the object's current owner.
- Principal Self (S-1-5-10):
  - A placeholder in an ACE on a user, group, or computer object in Active Directory.

(http://technet.microsoft.com/en-us/library/cc978401.aspx)

Registry Data types

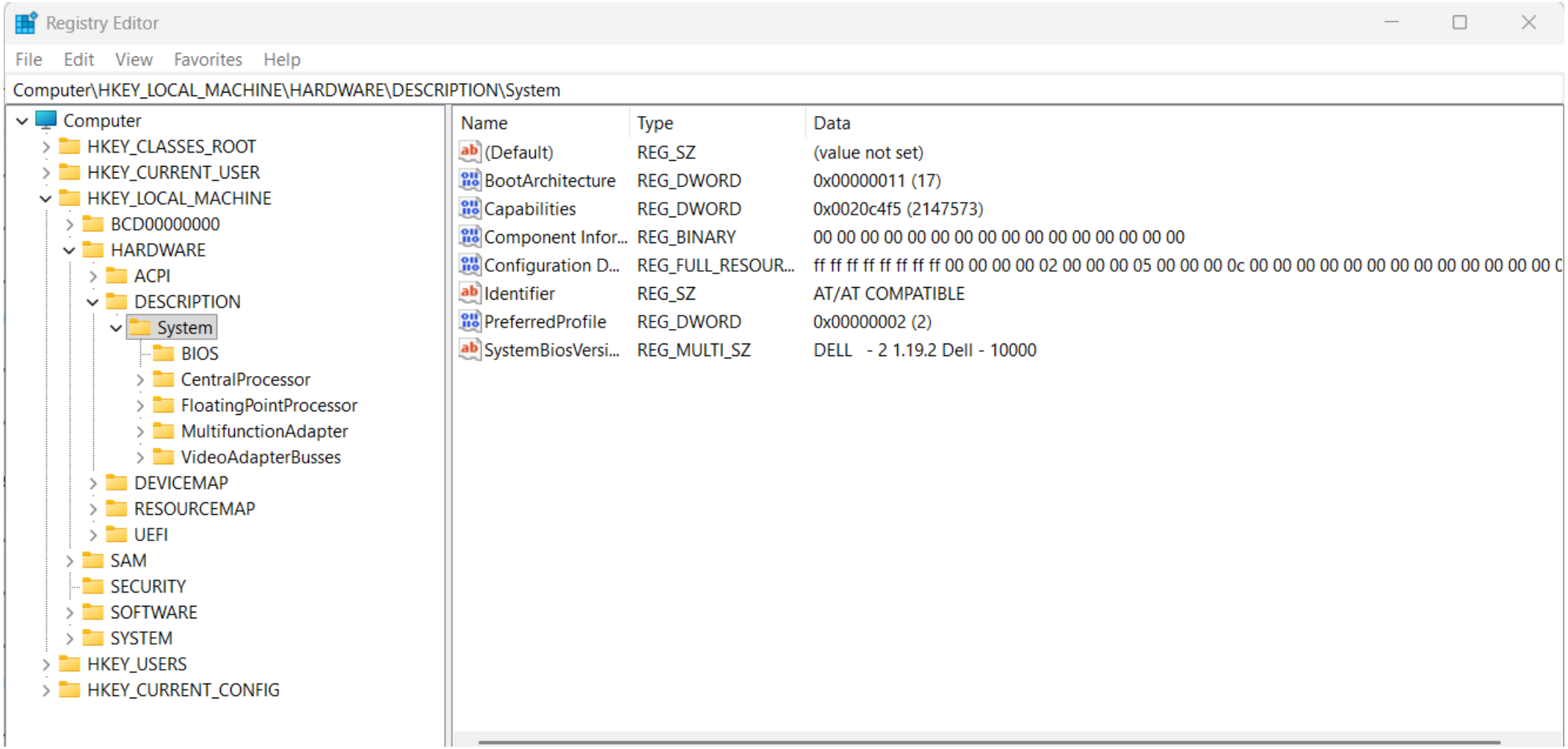
The following table lists the data types currently defined and used by the system.

Generally Readable

Data type	Description
REG_BINARY	Raw binary data. Most hardware component information is stored as binary data and is displayed in Registry Editor in hexadecimal format.
REG_DWORD	Data represented by a number that is 4 bytes long. Many parameters for device drivers and services are this type and are displayed in Registry Editor in binary, hexadecimal, or decimal format.
REG_EXPAND_SZ	A variable-length data string. This data type includes variables that are resolved when a program or service uses the data.
REG_MULTI_SZ	A multiple string. Values that contain lists or multiple values in a form that people can read are usually this type. Entries are separated by spaces, commas, or other marks.
REG_SZ	A fixed-length text string
REG_FULL_RESOURCE_DESCRIPTOR	A series of nested arrays designed to store a resource list for a hardware component or driver.

Require analysis

HKEY_LOCAL_MACHINE



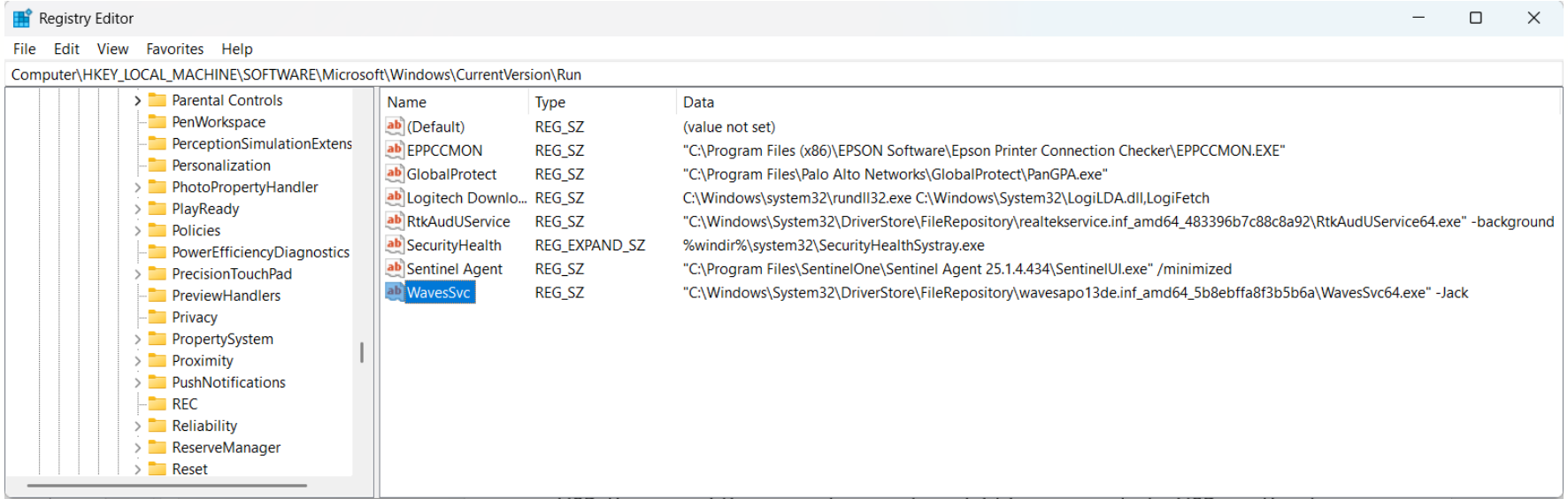
‘AutoRuns’

- If examining a case which may possibly concern malware inserting commands into an autostart location, is one way of executing malicious software.
  - If an unknown or suspect registry process is listed, then this should be examined.
- Is the evidence on the disk as a result of the actions of the user or a piece of malware? Has the system been ‘Trojaned’?

Also possible to check for malware tool entries in the registry – if you can find it in the file system – then a search using the same name or associated terms may find related material in the registry

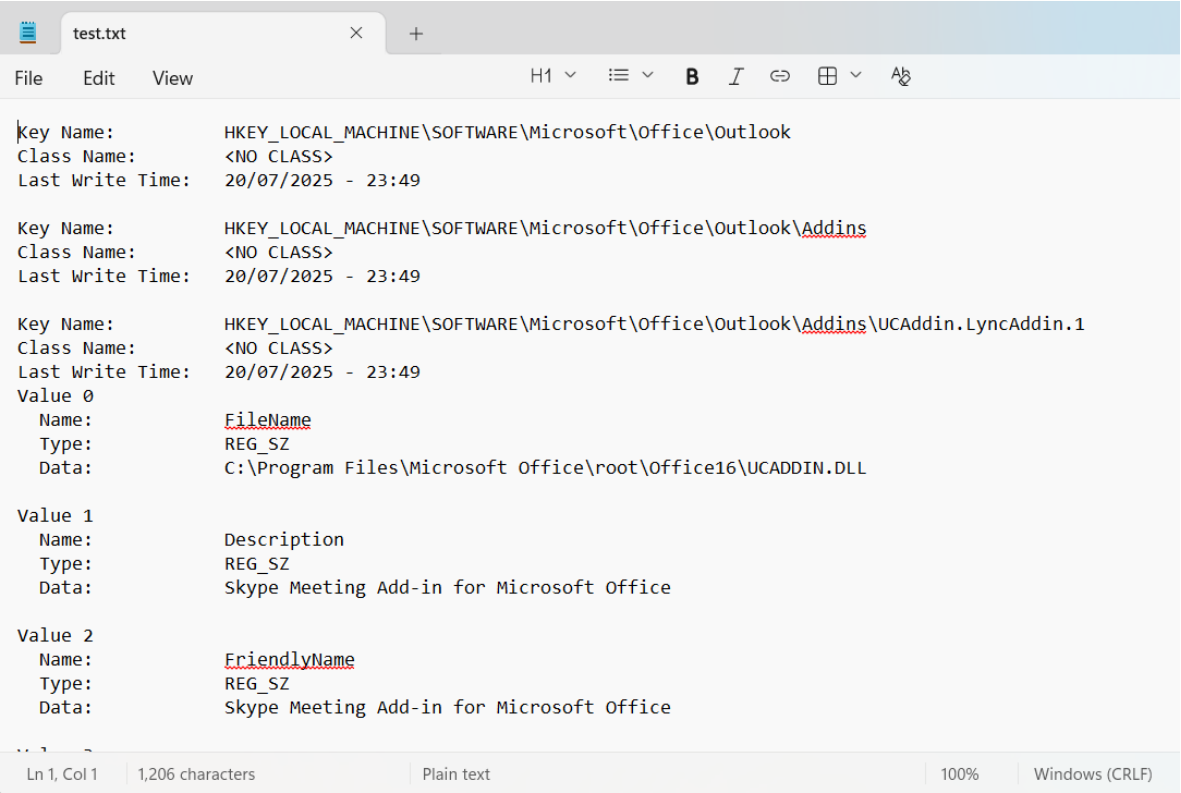
AutoRuns

- There are a number of places in the registry which can trigger programs to be executed under certain circumstances. Programs executed on boot up when the system starts are held in the following key:
  - **\HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\Run**
- Programs can also be executed when other actions occur including login, specific user actions etc.
  - \HKEY_LOCAL_MACHINE
  - \HKEY_CURRENT_USER
- Both contain a key, which causes an application to be launched whenever the command processor (cmd.exe.) is run
  - \SOFTWARE\Microsoft\command processor\autorun



Timestamps

- When a new Key / Value Name is added or a Value Name is changed or deleted the change is time stamped in the registry key.
- Timestamps can also be read using forensic analysis tools.

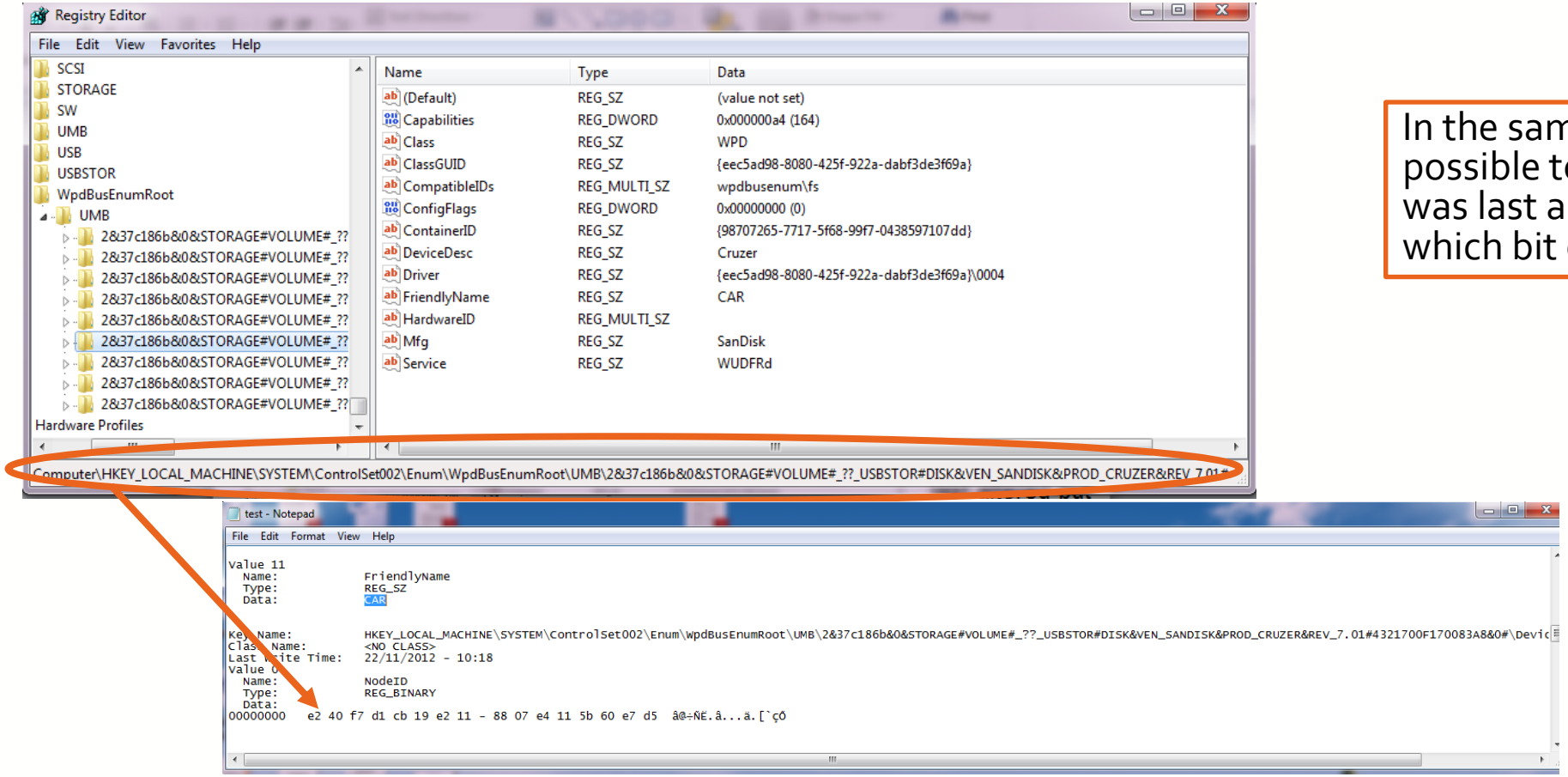


To view the Last Write Time for a registry key (and sub-keys):

1. Use REGEDIT.EXE to select the key, right-click it, and press Export.
2. Select a target location and save as a text file (.txt).
3. Open the file in notepad to display the times alongside each key /sub key

Timestamps

In the case of last write times it is possible to tell when the key was last written to but this may have a number of settings.

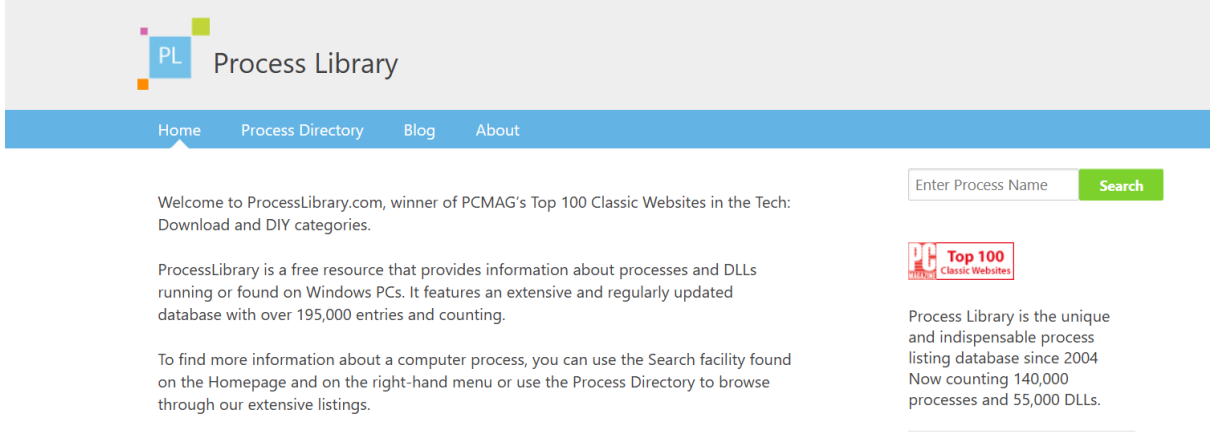


In the same way that it is possible to tell when a file was last altered but not which bit of the file.

Identifying a process

Processes identified in AutoRun can be examined to determine

- if they are part of the system processes,
- if they are part of an application
- if they represent a piece of malicious software.



See: - www.processlibrary.com

- Microsoft Autoruns

Limitations of Registry Evidence

- Does not record every action
- Timestamps may reflect system or application activity
- Keys can persist after uninstallation
- Interpretation requires corroboration with other artefacts

VSS and File History

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Windows Volume Shadow Copying

- The **Volume Shadow Copy Service** (VSS) allows **snapshots** to be taken of file systems
  - **Clone Shadow Copies:** A **split mirror** duplicates a volume by software or hardware mirroring, divided into a **live** and **mirror** volume – only the live volume is writable
  - **Copy-on-Write:** A **differential** copy only retains changes to the live data on the shadow volume, requiring overlaying with the live data to recreate an earlier state
- Only CoW is supported by the built-in VSS provider shipped with Windows
- Some applications (e.g., Exchange) are VSS-aware and can receive **freeze** and **thaw** messages to ensure internal consistency of backups

Forensic Value

- Preserves point-in-time snapshots of a volume
- May contain:
  - Deleted or overwritten files
  - Previous versions of documents
  - Historical Registry hives
  - Earlier system configurations
- Supports:
  - Timeline reconstruction
  - Before/after system comparisons
  - Recovery of pre-incident states
- Particularly valuable in:
  - Ransomware investigations
  - Insider activity cases
  - Post-incident forensic analysis

Remember:
- NTFS Feature
- Mounted as read-only drives for analysis

VSS limitations

- Snapshots may be deleted automatically
- 64 TB maximum volume size
- Common target of ransomware
  - Malware can disable it (or user)
- Not guaranteed to exist

System Restore

- Creates restore points of system files, drivers, registry, and installed applications
  - No longer enabled by default
- Allows rollback after problematic updates, drivers, or software installs
- Does not restore personal files
- Often disabled by default
- Must be manually enabled per drive
- Offers:
  - Registry rollback
  - Creates restore point artefacts.
  - Restore point snapshots under `System Volume Information`

Microsoft Windows File History

- From Windows 8 onwards, the equivalent of VSS has been offered by the **File History** (FH) service (and is more visible in Windows 10)
  - Internally this relies on similar differential/journaling structures as VSS but operates on files rather than blocks
  - Regular users may recover deleted or previous versions of files and folders
  - This allows forensic examiners to do the same, ideally with access to the (external?) drive on which backups are retained
- **File History requires the use of NTFS!**

- Low Priority for Forensics
  - A user-level backup feature
  - Periodically copies files (Documents, Desktop, etc.)
  - Contain older versions of user files
  - Preserve deleted or overwritten documents
  - Support timeline reconstruction of document edits

Microsoft Windows File History – The **$UsnJrnl**

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The change journal is retained in ***\$Extend\$UsnJrnl*** in an **alternate data stream** with entries containing

- Time of change
- Reason for change
- File/directory name
- File/directory attributes
- File/directory MFT record number
- File record number of the file's parent directory
- Security ID
- Update Sequence Number of the record
- Information about the source of the change

- Provides historical file activity
- Records events even after file deletion
- Supports timeline reconstruction
- Valuable in anti-forensics cases

Microsoft Windows File History – **$UsnJrnl Operation**

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The actual amount of storage taken up by File History, the frequency, and time horizon can be configured

- Within the alternate data stream, files are saved with an appended date/time stamp (resolution is that of NTFS: 100ns nominal)
- If external storage medium for journal backup is not available, a local cache is used (percentage to be occupied is configurable)
- On connecting the backup volume, the cache folder is moved to the backup volume and emptied

This offers a substantial degree of help for forensic investigators **if** no deliberate deletions or counter-forensic measures have been undertaken

Feature	System Restore	Volume Shadow Copy (VSS)	File History
Primary Purpose	Roll back system state	Snapshot of entire volume	Backup of user files
Present in	Windows 10 / Windows 11	Windows (NTFS volumes)	Windows 10 / 11
Scope	System files, registry, drivers, installed programs	Full volume snapshot (file-level access)	User libraries (Documents, Desktop, etc.)
Includes Registry?	Yes	Yes (if present at snapshot time)	No
Includes User Files?	No (not intended to)	Yes	Yes (targeted folders only)
Storage Location	System Volume Information	System Volume Information	External drive or network location
Created Automatically?	Sometimes (if enabled)	Yes (by OS events, updates, restore points)	Only if configured
User Visibility	Restore wizard	Hidden (requires tools)	Visible through UI
Forensic Value	Registry rollback evidence	Recover deleted/previous file versions	Recover user file history
Anti-Forensic Risk	Restore may mask changes	Shadow copies can be deleted	Limited (depends on config)

a transaction log is a file (i.e., log) of the communications (i.e., transactions) between a system and the users of that system (Jansen, 2006)

Log Analysis

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Event logs

- Record system, security, and application events
- Useful for timelines and incident reconstruction
- Can be deleted or overwritten
  - Recent files and jump lists
  - Browser artefacts
  - USB and device history

Log analysis in principle

- Monitoring logs allows managing of **user issues**, **security breaches**, **hardware failures**
- Log analysis tools offer:
  - automated parsing of fields,
  - geolocation lookups,
  - indexing,
  - generation of statistics and summaries
- Look for:
  - patterns based on statistics for requests, e.g. many POST requests
  - Many false positives
  - Much data irrelevant to your investigation

Skilled attackers have ways of avoiding detection

- disabling the collection of, or deleting or hiding this log evidence
- producing large quantities of noise events to purposely fill up the log data files prior to penetration
- modifying the computer operating system

Log Types and Analysis

- **Various log types**: Syslog, Web Server, Email logs, Antivirus logs, etc

- **Syslog Servers**
  - Standard for computer data logging
  - send diagnostic and monitoring data
  - offers a central repository for logs from multiple sources
  - automated events to trigger alerts
  - Logs may differ from one application to another – maintains standard per device logging
  - Offer alarms for events such as: Hardware errors, Application failures, Lost contact, Mis-configuration

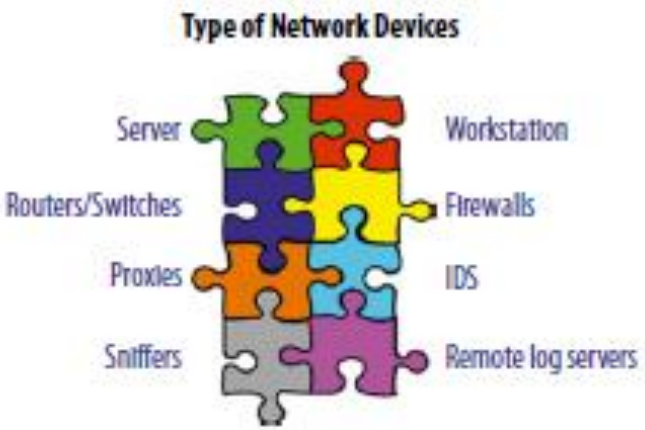
Log analysis in practise

- Normalise logs for analysis, e.g. use Splunk
- Create a list of keywords that are used to search evidence
- Always check timestamps

- **Event logs**: log-ins and log-outs, user command histories, root access events, ftp logs, log-on activity, files that users access, programs that users run, operations administrators perform
- **Application, Security, System event logs**: most common to check
- **Tools**:
  - Event Viewer
  - PSLogList
  - Log Parser
  - Plaso

Where to find event logs?

- Network routers
- Firewalls
- IDS
- Application servers (IIS, Netscape, Apache Web servers)
- Application proxies
- Application clients (Web browsers)
- Host system platforms (UNIX and Windows)



- General logging
  - time of day, identity of user, identity of files or objects being accessed, type of access or network traffic that is taking place, and IP of computer originating the traffic in the case of a network log.
- Specific logging
  - IDS is attack-specific namely,
  - an alert generated and logged by an IDS will identify that a particular attack has occurred and will indicate the vulnerability being exploited and possibly also the steps comprising the attack.

Windows Encryption Mechanisms

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Windows File and Volume Encryption

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1. Full disk encryption

- BitLocker: The BitLocker mechanism can provide volume encryption protected by a number of key material sources including USB tokens and built-in Trusted Platform Modules

2. File-level encryption

- EFS: The Encrypting File System provides a per-file encryption option; this does not include boot-up files, registry, or page files (inter alia)

3. Application-level encryption

- A number of third-party suppliers (e.g., BoxCryptor, tcplay) provide similar mechanisms for Windows

Caution!

Failing to recognise that a volume, files or folders are encrypted prior to imaging can have significant repercussions

Encryption as a Forensic Constraint

- Limits access to content at rest
- Metadata may still be available
- Live vs post-incident acquisition matters
- Timing in acquisition is important

<u>Encryption is not Evidence of Guilt</u>

- Often default OS behaviour
- May be organisational policy
- Context matters

Windows BitLocker

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BitLocker

- Although called BitLocker Drive Encryption, Microsoft's BitLocker is in fact a volume encryption mechanism
  - It requires the boot volume to be unencrypted (this secondary boot volume can be small, but must also be NTFS)
  - Key material can come from a number of sources, including USB pen drives, passwords or PIN codes, or a Trusted Platform Module (TPM)
    - Various combinations of TPM and other authentication mechanisms are also supported to provide multi-factor authentication
    - Windows 11 largely mandates use of a TPM 2.0; MDM systems can, e.g., also check if BitLocker and SecureBoot are supported – Windows 10 from Build 1607 and Server versions also allowed this with a TPM 1.2 module

Windows Device Encryption (1/2)

From Windows 8.1 onwards, encryption **may** be initiated automatically by default under the name **Device Encryption**

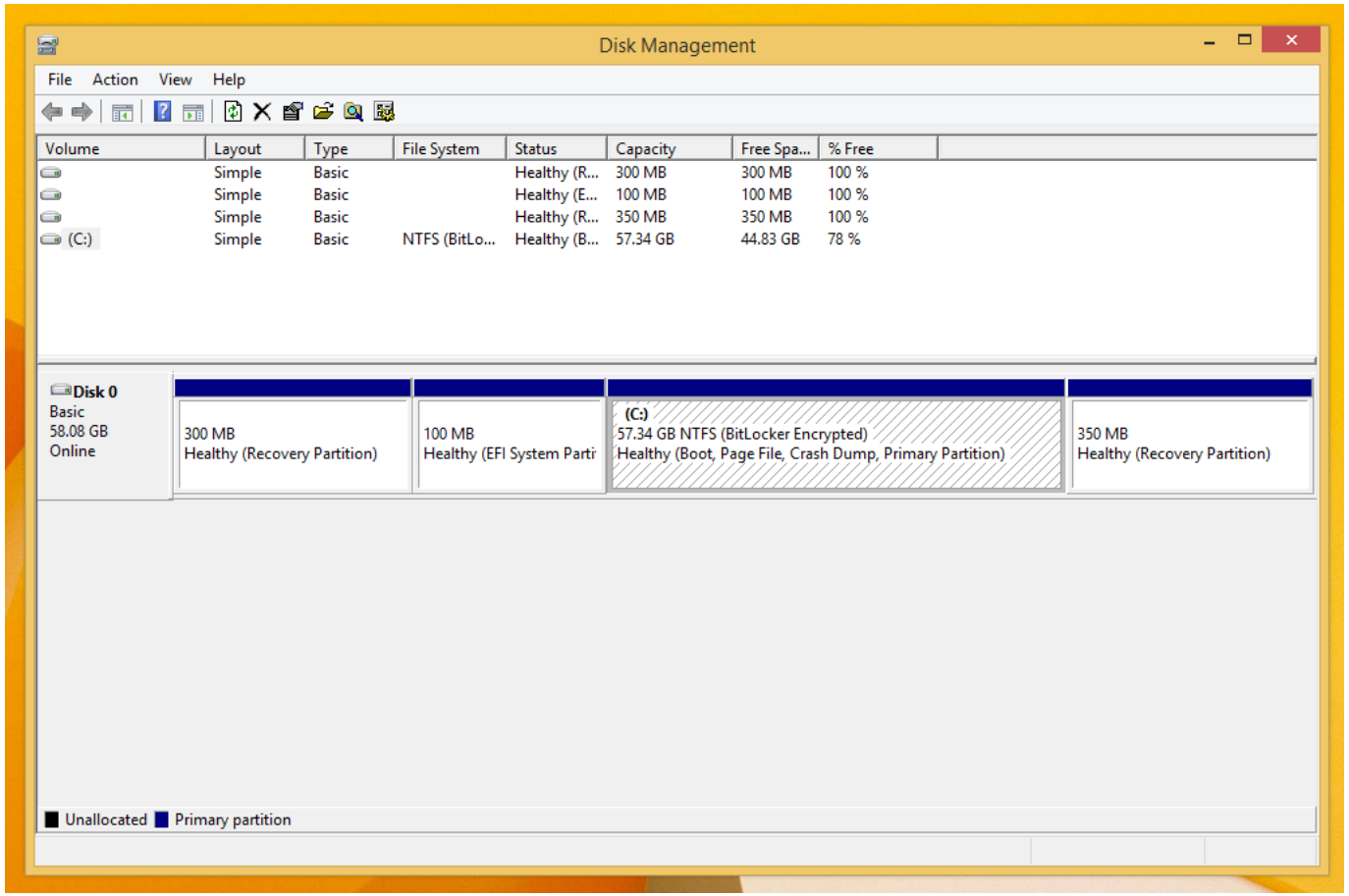
- becoming more important with platform unification in Windows 10 and 11 incorporating mobile devices
- In this case the Master Key is not protected (anyone with access to an administrative account may generate a **recovery key** linked to the Microsoft account linked to the device)
- Otherwise, an Active Directory group policy must be used to store the data within the Active Directory structure

Desktop systems generally do not support required features (including fixed, soldered-in memory, solid-state boot drive, and NDIS 6.30 support for all network interfaces)

- Some laptops may support this, but the feature seems mainly targeted to tablets and phones

Windows Device Encryption (2/2)

As can be seen, Device Encryption is largely a different name for BitLocker, but is not restricted to the Pro and Enterprise tiers (SKUs) any longer:



The challenge

- **BitLocker**
  - Full disk encryption built into Windows
  - Protects data **at rest**
  - Commonly enabled by default on modern systems
- **Forensic implications**
  - Disk content unreadable when powered off
  - File system metadata may still be identifiable
  - Live system state is critical for access

Windows EFS

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Encrypting File System

Encrypts **individual files**

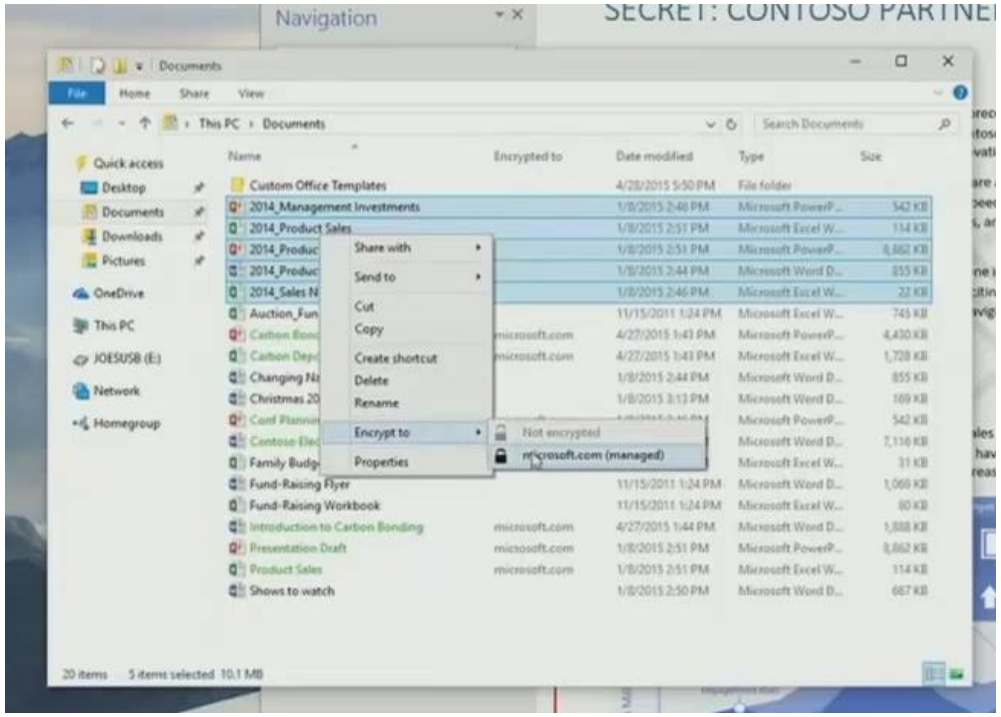
Encryption tied to **user credentials**

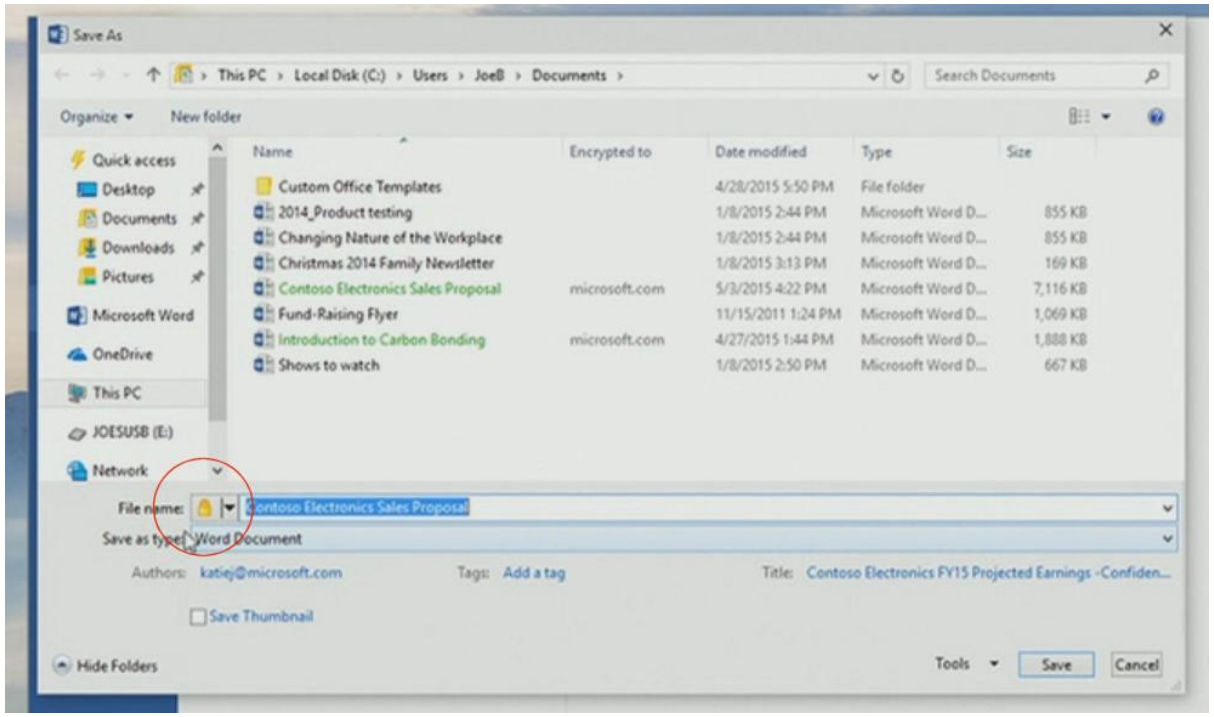
Legacy feature, rarely used in enterprise environments

- The EFS mechanism operates as a file system filter and was introduced in V3.0 of NTFS in 1999
- Files are encrypted individually with a symmetric cipher
- This (pseudo-)randomly generated key (File Encryption Key or FEK) is then encrypted using a public key cryptosystem and stored alongside the encrypted file in the $EFS alternative data stream
- On reading the file, the certificate associated with the FEK is retrieved, and the FEK is decrypted with the matching private key
- EFS operates on a per-file basis, but an "Encryption" attribute can be set on a folder level and is inherited
- Files and directories copied within an NTFS volume remain encrypted

EFS User Interface

The EFS user interface has long been somewhat debatable





EFS Limitations (1/2)

EFS has several limitations arising from its design:

- Key Management: Security of EFS-encrypted files is typically not limited by the strength of the bulk cipher, but by management of public keys including **recovery keys**
- If a **recovery agent account** is compromised, this gives access to all EFS encrypted files

- File System Transitions: Copying files to non-NTFS volumes such as FAT or network file systems results in files being decrypted
- Such transitions may be transparent to users, resulting in storage in plain text
- File systems with **links** to others (e.g., folders) in them are problematic

EFS Limitations (2/2)

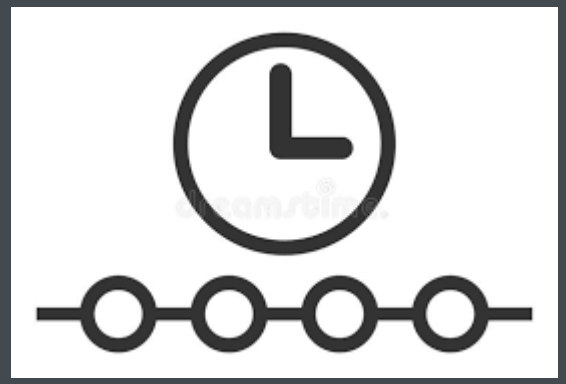
- EFS is mainly intended to protect **data at rest**
  - i.e., files on storage media during times when the operating system is not running
- When plain-text files are encrypted, this may not happen **in situ**:
  - **Remnants of the original file may still be found in the file system's data structures**
- The contents of decrypted files may propagate to several unencrypted locations
  - Paging and printer spooling subsystem, hibernation files, but also various temporary files created by applications
- EFS is purely a client-side mechanism; while some key management issues can,
  - e.g., be addressed using smart cards for key storage, it may be possible to recover plain-text data from other loci

The verdict

- BitLocker and EFS can technically coexist
- In practice, EFS is rarely used or offered when BitLocker is enabled
- Modern Windows environments strongly discourage EFS when BitLocker is active

- EFS adds **no meaningful extra security**
- It increases **key management complexity**
- It creates **recovery and support risks**

- As a result:
  - EFS is often **disabled by policy**
  - The UI option may not appear
  - Organisations actively avoid combining them



Timeline Analysis

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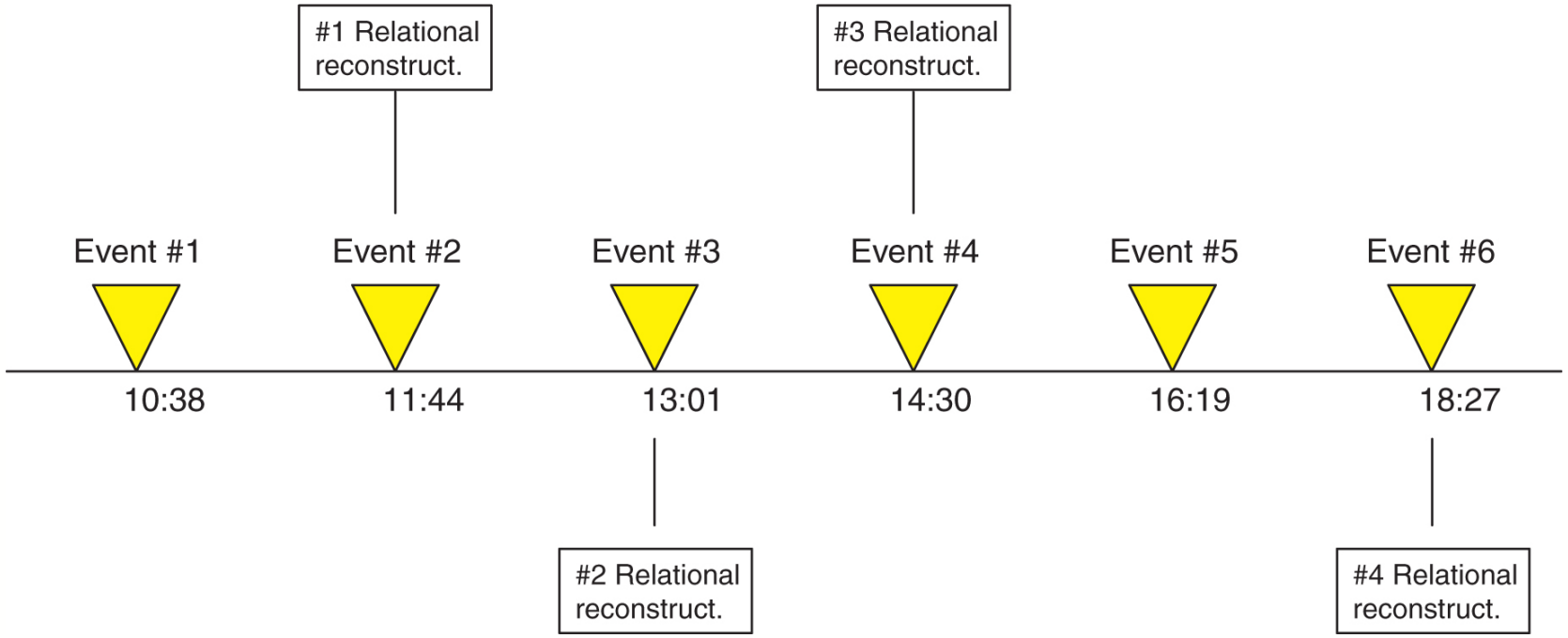
The timeline

- The outline of the incident
  - Aim is to identify digital timestamps
- A linear event analysis to establish
  - The sequence of events
  - Any related events



Timeline

- The outline of the incident
  - Aim is to identify digital timestamps
- A linear event analysis to establish
  - The sequence of events
  - Any related events
- Constraints
  - Computer clock accuracy
  - Time zones
  - Manipulation of timestamps/ Timestomp

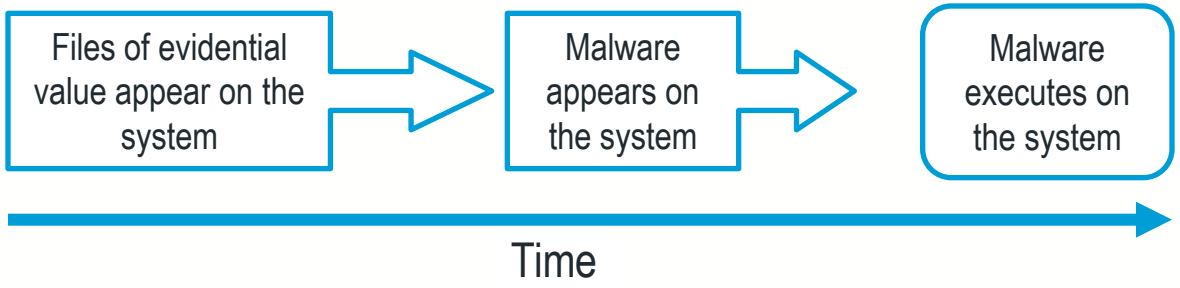
Time-Lining

- Ordering events of interest in time
  - intention of obtaining an overview of the sequence of events
  - an insight into cause-and-effect relationships and motive.
- Evidence timestamps

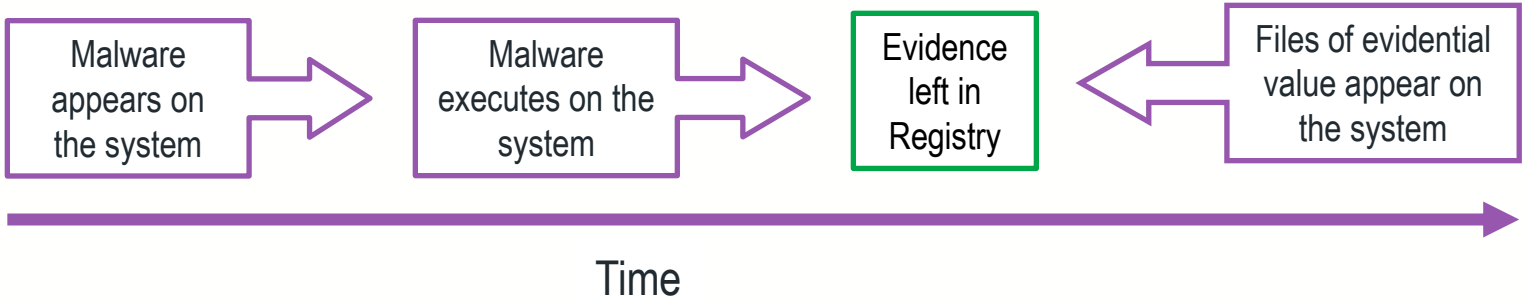
1. Log-on and log-off events;
2. Internet browsing site location and content and chat room activities;
3. Application usage (file transfer and software usage);
4. E-mail communication events;
5. File usage events (e.g., creating, modifying, and accessing documents);
6. Router log events.

Examining the Timeline

IF the malicious software arrived on the system ***after*** the files of evidential value appeared, then the malware may have had no effect.

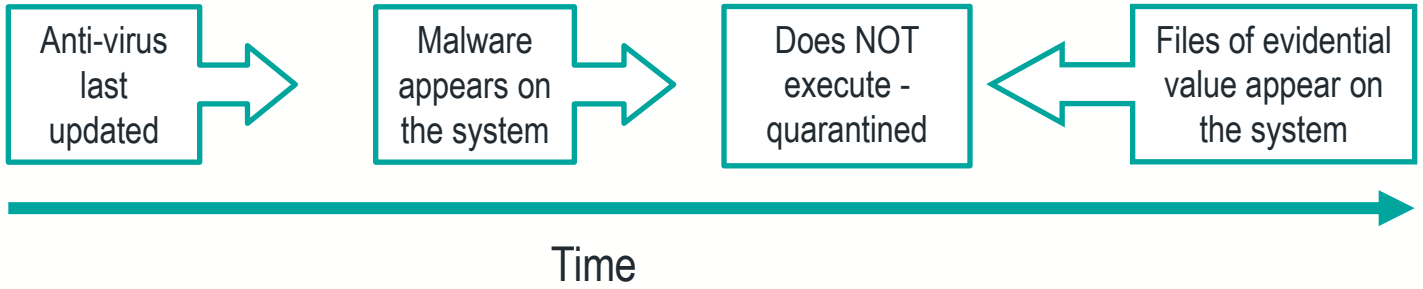


IF the malware impacts on the case then it should be examined to determine if it has executed.

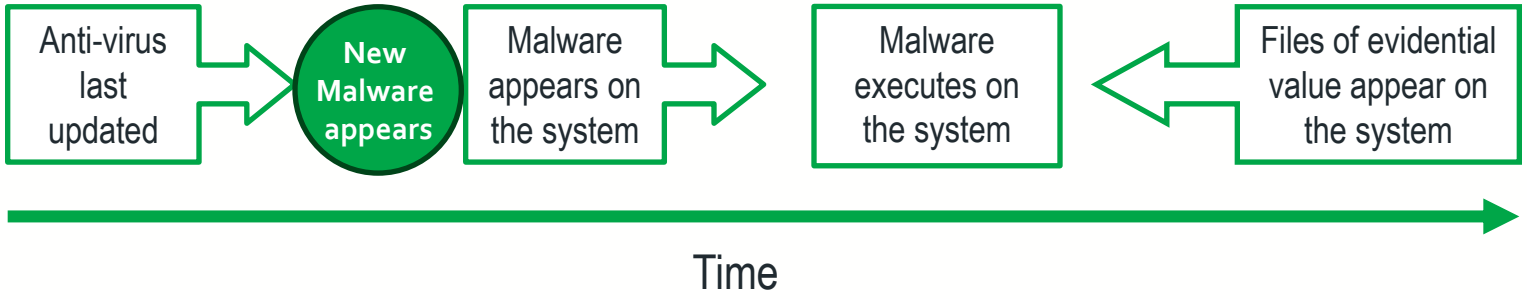


Examining the Timeline (2)

IF antivirus is present what is the anti-virus update history?



IF the user missed an update?



Time-Lining Issues

- Identifying time-related patterns (e.g. MAC timestamps);
- Cross correlating different time sequences;
- Correlating these sequences with other physical evidence.

- Time zones or relative time
- Clock synchronisation

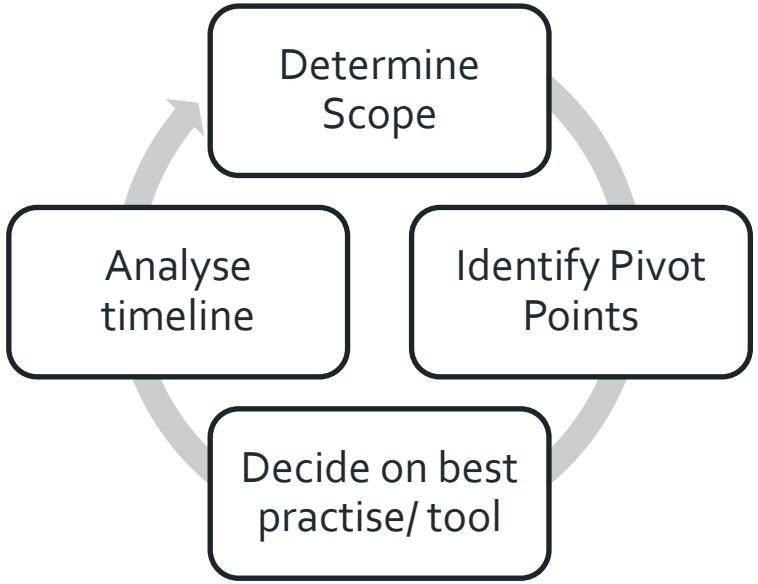
Analysis

Pivot Point: Examine what happened immediately before and after the incident

- Time of incident
- Network/ Process Activity
- Suspicious files and file types
- Specific activities

Consider:

- Last time access
- LNK files
- Focus on the questions that need to be answered



Tools:

-Log2timeline - https://github.com/log2timeline/plaso

-Tapestry - https://github.com/derekedw/Timeline

Analysis Guidance

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Filesystem Data

- Filesystem metadata - $MFT, $Secure, $STANDARD_INFORMATION attributes within the MFT
- Directories
- Files: Deleted, Unallocated metadata
- Timestamps: Modified, Accessed, Creation, Change

Windows Artifacts

- User communications: emails, calendars, chat, messengers
- File downloads and sources: email, file sharers, chat
- File creation/ opening
- External devices
- Physical location: WiFi SSID, Time Zone, cookies
- Browser usage, etc.

Registry Keys

- 'Most Recently Used' (MRU) keys: HKCU\Software\Microsoft\Office\12.0\Word\File,
- HKCU\Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\LastVisitedMRU

Questions?

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Summary

- Windows leaves rich forensic traces
- Evidence must be correlated across artefacts
- Interpretation must be cautious and defensible

Useful sources

- https://www.file-extensions.org/
- https://www.garykessler.net/library/file_sigs.html
- Brett Shavers, John Bair, 2016. Hiding Behind the Keyboard, Chapter 7 - Antiforensics, Syngress, Pages 153-172, ISBN 9780128033401

References

Chris Prosise, Kevin Mandia, and Matt Pepe. 2014. *Incident Response & Computer Forensics*, 3rd Ed. McGraw-Hill, Inc., USA.

Zhu Y., Gladyshev P., James J. (2009) Temporal Analysis of Windows MRU Registry Keys. In: Peterson G., Shenoi S. (eds) Advances in Digital Forensics V. DigitalForensics 2009. IFIP Advances in Information and Communication Technology, vol 306. Springer, Berlin

Eoghan Casey, Curtis W. Rose, Handbook of Digital Forensics and Investigation, 2010, Heidelberg. https://doi.org/10.1007/978-3-642-04155-6_6

Harlan Carvey, in Windows Forensic Analysis Toolkit (Fourth Edition), 2014

Syslog Tutorial: How It Works, Examples, Best Practices, and More